General Crop Disease Management Guidelines (Starter Reference)

IMPORTANT: This is an original starter document written for the AgriSense AI knowledge base, summarizing general Integrated Pest Management (IPM) principles. It is NOT a reproduction of any ICAR or government publication. Before production deployment, replace or supplement this file with actual downloaded PDFs from icar.org.in and your State Agriculture Department, then re-run `scripts/ingest_knowledge_base.py`.

1. Principles of Integrated Disease Management

Effective crop disease management rests on four pillars, applied together rather than in isolation:

1. **Prevention** — resistant varieties, certified disease-free seed, crop rotation, field sanitation, and balanced nutrition reduce the chance of an outbreak starting at all.
2. **Monitoring** — regular field scouting (at least twice a week during the growing season) catches disease at the earliest, most treatable stage.
3. **Timely Intervention** — matching the treatment (chemical, biological, or cultural) to the disease's spread rate. Fast-moving diseases like late blight need action within 24-48 hours; slower diseases tolerate a few days' delay.
4. **Resistance Management** — rotating fungicide/pesticide chemical classes between sprays prevents pathogens and pests from evolving resistance, which protects the long-term effectiveness of every product in the toolkit.

2. Reading a Pesticide Label

Every registered pesticide label carries five pieces of information a farmer must check before spraying:

- **Active ingredient and concentration** (e.g. Mancozeb 75% WP)

- **Recommended dose** per litre of water
- **Target pest/disease** the product is registered for
- **Pre-harvest interval (PHI)** — minimum days between last spray and harvest
- **Safety precautions** — protective equipment, re-entry interval, and storage instructions

3. Safe Spraying Practices

- Spray during early morning or late evening when temperatures are cooler and wind is calm, to reduce drift and evaporation.
- Wear protective gloves and a mask; wash hands and equipment thoroughly after spraying.
- Never spray directly before rain — check a short-range weather forecast first.
- Keep children and animals away from freshly sprayed fields until the product has dried.
- Store unused pesticide in its original, labeled container, away from food and out of children's reach.

4. Weather and Disease Risk

Fungal diseases (late blight, early blight, powdery mildew, leaf spots) generally thrive in specific weather windows:

- **High humidity + moderate temperature (15-25°C) + leaf wetness** → favors most fungal leaf diseases, especially late blight.
- **Hot, dry conditions** → favors powdery mildew and spider mite outbreaks (mites, not a disease, but often confused with one).
- **Extended dry spells followed by heavy rain** → can trigger rapid bacterial spot and bacterial spot-like outbreaks as bacteria splash from soil onto lower leaves.

A simple rule of thumb: if 3+ consecutive days combine high humidity with overnight temperatures above 15°C, treat this as a high-risk window for fungal disease and consider a preventive spray even before symptoms

appear, especially on susceptible crops like potato and tomato.

5. When to Escalate Beyond Home Treatment

Contact your nearest Krishi Vigyan Kendra (KVK) or agriculture extension officer if:

- A disease spreads to more than 25% of the field despite treatment
- You cannot identify the disease with reasonable confidence
- Symptoms suggest a viral disease (no chemical cure exists — early removal and vector control are the only options)
- You are unsure which pesticide is registered/legal for your crop and region

6. Government Support Schemes (verify current details locally)

Farmers dealing with crop loss from disease outbreaks may be eligible for support under national and state schemes such as the Pradhan Mantri Fasal Bima Yojana (crop insurance) and state-specific input subsidy programs. Claim windows and eligibility criteria change over time — always confirm current details with your local agriculture office rather than relying on this document.